Carbon-Aware Load Shifting Across Multiple Buses: Increasing Accuracy and Impact through Validity-Constrained Shifting

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Abstract

An increasing number of flexible electric loads are interested in reducing their carbon footprint through carbon-aware load shifting. A frequently used carbon emission metric is Locational Marginal Carbon Emissions (LMCEs). However, since LMCEs are a local sensitivity metric, load shifts may be ineffective or counterproductive when they are too large, making it useful to know the validity range of the LMCE, i.e. the range of load shifts for which the LMCE value remains constant. Prior work defining the LMCE validity range only considers single-node load shifts, and terminates at the first change in the set of binding power flow constraints (i.e., terminating at the first basis pivot), even when the LMCE value itself remains unchanged. In this paper, we first improve the LMCE validity calculation to consider multi-node load perturbations, giving rise to a validity region rather than a validity range, and use an iterative procedure that extends the LMCE validity region beyond the first basis pivot if the LMCE does not change. Next, we demonstrate the practical value of LMCE validity regions by embedding them as constraints in a spatio-temporal load-shifting optimization problem. In a year-long load-shifting case study on the RTS-GMLC system, we demonstrate the practical importance of considering LMCE validity regions that are accurate, yet not overly conservative. The proposed iterative method achieves 26–35 kt CO_2 *(11–14%) more in annual emissions reductions than the more conservative approach based only on OPF basis pivots, outperforming it on over 93% of operating days. Further, the proposed approach achieves a negligible prediction error below 0.015%. By contrast, unconstrained load shifting (which ignores validity regions entirely) incurs prediction errors of 50–69% of reported reductions, rendering its apparent emissions savings largely unreliable.*

Keywords: Locational marginal emissions, load shifting, carbon-aware optimization.

1. Introduction

Emissions from electricity generation constitute a major share of total greenhouse gas emissions [1]. However, increasing renewable penetration makes the emissions impact of electricity consumption highly time and location dependent [2], providing an opportunity for flexible electric loads to reduce emissions by shifting electricity consumption across time and location to align with low-emission generation. To guide emission-aware load shifting efforts, consumers leverage carbon signals, metrics that capture the temporal and spatial emissions intensity of electricity consumption. Example applications include industrial processes [3], cloud computing workloads [4], electrolyzers [5], and demand-side flexibility in distribution networks using locational marginal emissions signals [6].

Multiple methods have been proposed for designing accurate and effective carbon signals [7]. The most widely used metric is Average Carbon Emissions (ACE), which is endorsed by the Greenhouse Gas Protocol and is widely available [8]. However, as a locationally aggregated, backward-looking measure, ACE often misrepresents the emissions impact of demand changes (e.g. load shifting) [9, 10, 4, 11]. For load-shifting applications, a more appropriate signal is Locational Marginal Carbon Emissions (LMCEs), which provide nodal-level resolution. LMCEs quantify the sensitivity of a power system's emissions to a change in load at a target bus (node), capturing the re-dispatch caused by marginal changes in load [12, 13]. Analogous to Locational Marginal Prices (LMPs), they quantify the emissions impact of a marginal load change rather than its economic cost and can be negative when added load relieves congestion and allows substitution toward lower-emission generation. Additional advances have expanded LMCE methodology, including dynamic formulations [14], distribution-level extensions [6], and practical load-shifting applications [15]. A comparative analysis of carbon intensity signals [11] suggests that LMCE is the most effective metric to achieve short-term reductions in carbon emissions from grid operations.

Because LMCEs are local sensitivity factors, they are only reliable for sufficiently small load changes, which we refer to as the *LMCE validity region*. Outside this region, LMCE-guided load-shifting decisions may produce substantial prediction errors, causing realized emissions impacts to differ from predicted emissions savings [15]. For practical load shifting applications, it is therefore valuable to assess *how large* the LMCE validity range is, answering the question of "how much load can we shift before the LMCE metric loses its accuracy?" Prior work [16] addressed this question using sensitivity analysis of the optimal power flow (OPF) solution. Specifically, [16] defines the LMCE validity range for load perturbations at an individual bus as the amount of change that is possible before the set of binding constraints at the optimal solution (i.e. the OPF basis) changes. They further showed that the validity ranges are often small, particularly under congestion, which suggests that the practical value of LMCE-guided load shifting may be small.

While the method in [16] represents important progress in defining the LMCE validity region, it has two important shortcomings. First, it derives validity ranges only for single-bus load perturbations despite the fact that many practical demand-side flexibility applications involve multiple flexible loads, such as industrial processes, data centers, or electrolyzer fleets, adjusting their demand simultaneously across different locations in the network [15, 4, 5]. The LMCE validity region fundamentally changes when multiple loads shift simultaneously, and simultaneous shifts cannot be accurately represented through independent single-bus validity limits. Consequently, LMCE-guided load shifting across multiple buses requires *validity regions* defined over joint load perturbations, rather than *validity ranges* defined for a single bus load perturbation.

Second, [16] defines LMCE validity only based on changes in the OPF basis, i.e. it terminates the validity range as soon as the basis changes. This definition is motivated by the fact that a basis change can cause the marginal generators to change, and thus cause a change in the LMCE value. However, LMCE values can remain unchanged despite changes in the OPF basis when multiple generators have identical emissions rates, a situation that frequently arises during periods of high renewable generation. Consequently, a change in the basis does not necessarily imply that the LMCE changes. We observe that the practical effect of this distinction is particularly consequential for load shifting across multiple buses, where validity regions can become overly restrictive. As a result, analytical methods may substantially underestimate the true LMCE validity region and unnecessarily constrain carbon-aware load shifting.

To address the above limitations, we derive an improved method to compute LMCE validity regions for multi-bus load perturbations and propose an iterative procedure to extend the LMCE validity region beyond the first basis pivot if the LMCE does not change. Further, we demonstrate how LMCE validity regions may be integrated in a practical carbon-aware load shifting application.

1.1. Contributions

This paper develops and tests a framework for operationally reliable, simultaneous LMCE-guided load shifting using extended multi-bus validity regions. The contributions are:

1. **Multi-dimensional validity regions for coordinated load shifting.** We propose a method to derive analytical LMCE validity regions for simultaneous load shifting across multiple buses, where each analytical LMCE validity region takes the shape of a polytope in the load shifting space (as opposed to a one-dimensional validity range for individual buses in [16]). We then introduce an iterative method that identifies non-consequential basis pivots to recover larger LMCE validity regions than purely analytical approaches. These extended regions are defined as the union of multiple polytopes.

2. **Validity-constrained optimized load shifting.** We incorporate the LMCE validity regions directly into a spatio-temporal load-shifting optimization, using a convex hull approximation of the LMCE validity region to obtain a convex set of constraints. We demonstrate through a case study that validity-constrained load shifting yields decisions that preserve prediction accuracy while maximizing emissions reductions.

2. LMCE Validity Range Computation

In this section, we propose a method to compute LMCE validity regions. We build upon the method in [16], but evolve it to derive regions for simultaneous multi-node load shifts (as opposed to ranges for individual nodes) and propose a new iterative procedure that yields larger validity regions. Note that our formulation is based on a standard DC OPF formulation. While other OPF formulations (including, e.g., AC power flow constraints or unit commitment variables) could be theoretically interesting to consider, most carbon-aware load shifting seeks to impact the dispatch of generation in the real-time electricity market clearing. This setting is most accurately reflected by a DC OPF formulation.

2.1. LMCE Computation

To frame our discussion of LMCE validity regions, we first describe how LMCE values are computed for a given operating point.

Optimal Power Flow Formulation We start from a standard DC-OPF formulation, which minimizes generation cost subject to power balance, generator capacity, line flow limit, and reference angle constraints:

$$\begin{aligned} \min_{P,\theta} \quad & c^\top P \\ \text{s.t.} \quad & J_P P + \Psi\theta = D, \quad \underline{P} \le P \le \overline{P}, \\ & -\Omega \le \Phi\theta \le \Omega, \quad \theta_{\text{ref}} = 0. \end{aligned} \tag{1}$$

Here, G and N denote the sets of generators and nodes. The variables $P \in \mathbb{R}^{|G|}$ and $\theta \in \mathbb{R}^{|N|}$ are the generator dispatch and voltage angles, J_P is the generator-to-bus incidence matrix, Ψ is the bus susceptance matrix, Φ maps angles to line flows, and D is the demand vector. For the basis sensitivity analysis, we convert (1) to standard equality form by introducing slack variables for the generator and line inequalities. Solving this slack-transformed formulation via the simplex method yields a partition of the variables into *basic variables* that are between their bounds at the optimum and *non-basic variables* that are at their minimum or maximum bounds [17]. The active constraints at the optimum (i.e. all equality constraints and inequality constraints whose slack variables are zero) define a square and invertible basis matrix $B^\star$, which governs how the solution, and in particular the value of the basic variables, respond to perturbations in D. Specifically, the optimal values of the basic variables x_B, including generator outputs, voltage angles, and slack variables not at a bound, are given by $x_B = (B^\star)^{-1}b^\star$, where $b^\star$ is the set of rows of the right-hand side vector associated with the active constraints.

LMCE Computation LMCEs quantify the sensitivity of total system emissions to a marginal load increase at a target bus. Physically, this corresponds to the change in generation required to balance an additional 1 MW of load while the active constraints remain unchanged.

To compute the LMCE at node $n \in N$, we represent the 1 MW demand increase as a perturbation to the right-hand side of the active constraints. Specifically, we define a perturbation vector $r \in \mathbb{R}^m$, where m is the number of active constraints at the optimum (equivalently, the number of basic variables), with a single nonzero entry of 1 in the row corresponding to the power-balance constraint at node n, and zero elsewhere. Since non-basic variables are at their bounds, only the basic variables adjust to maintain feasibility. Their response is obtained by solving

$$B^\star \Delta x_B = r \implies \Delta x_B = (B^\star)^{-1} r, \tag{2}$$

where $\Delta x_B = (\Delta P_B, \Delta\theta_B, \Delta s_B)^\top$ collects the changes in basic generator outputs, voltage angles, and slack variables per 1 MW increase in demand at node n. The corresponding change in generator dispatch is extracted via

$$\frac{\partial P}{\partial D_n} = S_P \, \Delta x_B, \tag{3}$$

where $S_P \in \{0,1\}^{|G|\times|x_B|}$ is a selection matrix that extracts the generator-output components of Δx_B, with one row per generator and a single nonzero entry in each row at the column corresponding to that generator's position in x_B. The LMCE at bus n is then the inner product of this dispatch response with the vector of generator emission rates $e \in \mathbb{R}^{|G|}$ (tCO_2/MWh):

$$\text{LMCE}_n = \frac{\partial E}{\partial D_n} = e^\top \frac{\partial P}{\partial D_n} = e^\top S_P \, \Delta x_B, \tag{4}$$

where E is the total system emissions.

2.2. LMCE Validity Regions

Given the LMCE at a specific operating point, we derive the LMCE validity region, defined as the set of simultaneous load shifts for which the LMCE values remain unchanged. We consider a set of flexible nodes $\mathcal{F} \subseteq N$, where N is the set of all nodes in the network. The vector $\Delta D \in \mathbb{R}^{|\mathcal{F}|}$ represents the joint load shift, where ΔD_n denotes the change in demand at node $n \in \mathcal{F}$, with positive values corresponding to demand increases and negative values corresponding to demand reductions. For example, with two flexible nodes $\mathcal{F} = \{n_1, n_2\}$, the vector $\Delta D = (+50, -50)$ MW represents a 50 MW load increase at n_1 and a simultaneous 50 MW reduction at n_2. We seek to characterize all joint load shifts ΔD for which the LMCE values remain unchanged, which defines a validity region in $\mathbb{R}^{|\mathcal{F}|}$. We first derive an analytical validity region based on the OPF basis, which by construction guarantees that the LMCE remains the same, before introducing an iterative procedure that recovers a larger region by checking directly for changes in the LMCE.

2.2.1. Analytical Validity Regions: For each flexible node $n \in \mathcal{F}$, define $r_n \in \mathbb{R}^m$ as the unit perturbation vector with a single nonzero entry of 1 at the row corresponding to node n's power balance constraint and zero elsewhere. Collect these vectors as columns of the matrix $R = [r_{n_1}, \ldots, r_{n_{|\mathcal{F}|}}] \in \mathbb{R}^{m\times|\mathcal{F}|}$, and let $\Delta D \in \mathbb{R}^{|\mathcal{F}|}$ denote the vector of simultaneous load shifts, where entry ΔD_n is the change in demand at node $n \in \mathcal{F}$. Since the basic variables respond linearly to right-hand side perturbations, the combined response to simultaneous load shifts across all flexible nodes is

$$\Delta x_B = (B^\star)^{-1} R\,\Delta D, \tag{5}$$

where $R\,\Delta D \in \mathbb{R}^m$ is the total perturbation to the right-hand side of the active constraints. Within the fixed basis, each basic variable x_i then evolves as

$$x_i = x_i^* + \left[(B^\star)^{-1}R\right]_i \Delta D, \tag{6}$$

where x_i^* is the optimal value of x_i at the base operating point and $[(B^\star)^{-1}R]_i$ is the i-th row of $(B^\star)^{-1}R$. To maintain the current OPF basis and ensure the LMCE remains unchanged, x_i must remain within its lower and upper bounds $[\ell_i, u_i]$, determined by generation capacity limits, transmission limits, and positivity of slack variables. Substituting the linear response into $\ell_i \le x_i \le u_i$ and rearranging gives one upper and one lower half-plane constraint on ΔD per basic variable:

$$\left[(B^\star)^{-1}R\right]_i \Delta D \le u_i - x_i^*, \tag{7}$$

$$-\left[(B^\star)^{-1}R\right]_i \Delta D \le x_i^* - \ell_i. \tag{8}$$

The analytical validity region $\mathcal{V}^{\mathrm{A}}$ is the intersection of all half-plane constraints of the form (7) and (8) over all basic variables i, taking the tightest bound in each direction to collectively ensure that no basic variable exceeds its bounds for any simultaneous load shift ΔD:

$$\mathcal{V}^{\mathrm{A}} = \left\{ \Delta D : \begin{array}{c} (B^\star)^{-1}R\,\Delta D \le u - x_B^*, \\ -\,(B^\star)^{-1}R\,\Delta D \le x_B^* - \ell \end{array} \right\}, \tag{9}$$

where $u, \ell \in \mathbb{R}^m$ are the vectors of upper and lower bounds on the basic variables. This defines a convex polytope in $\mathbb{R}^{|\mathcal{F}|}$. By construction, $\mathcal{V}^{\mathrm{A}}$ is a conservative inner approximation of the true LMCE-invariant region.

Note that the scalar validity range for single bus perturbations defined in [16] can be recovered by restricting the load shifts as $\Delta D = \lambda v$. Here, $v \in \mathbb{R}^{|\mathcal{F}|}$ is a direction vector defining the relative participation of each flexible node, normalized such that $\sum_{n\in\mathcal{F}} |v_n| = 1$, and $\lambda \in \mathbb{R}$ is a scalar representing the total magnitude of the load shift. This demonstrates that the scalar validity ranges from prior work are a special case of the more flexible validity regions derived here.

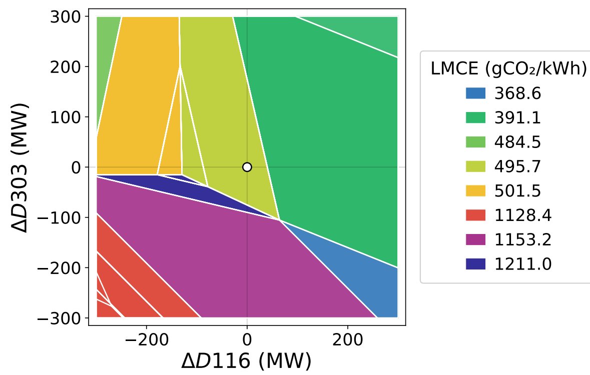

Figure 1: OPF basis regions in the two-bus load shift space spanned by buses 116 and 303 of the RTS-GMLC system at hour 20 of 2020/01/02. Each colored region corresponds to a distinct OPF basis with a constant LMCE value, shown in the legend. The white dot marks the baseline operating point.

2.2.2. Iterative Validity Region The analytical validity region $\mathcal{V}^{\mathrm{A}}$ guarantees that the LMCE remains unchanged by ensuring the OPF basis does not change, but an unchanged basis is only a sufficient condition for an unchanged LMCE, not a necessary one. In particular, when successive marginal generators share identical emissions rates, a situation commonly encountered during periods of high renewable generation, the LMCE remains unchanged even though the OPF basis changes. As a result, $\mathcal{V}^{\mathrm{A}}$ may be smaller than the true set of load shifts for which the LMCE remains unchanged.

To recover a larger region over which the LMCE remains unchanged, we explore neighboring OPF basis regions adjacent to $\mathcal{V}^{\mathrm{A}}$ and check whether the LMCE remains unchanged there. As illustrated in Figure 1, the load shift space partitions into basis regions, where same-colored neighbors have identical LMCE despite different active constraints. These non-consequential basis changes are exactly what the iterative method extends across. Each neighboring basis region is itself a convex polytope $\mathcal{P}_k$ in $\mathbb{R}^{|\mathcal{F}|}$, and is accessible by crossing one of the facets of the current polytope. For each facet of $\mathcal{V}^{\mathrm{A}}$, we evaluate the LMCE at a test point slightly beyond the facet,

$$\Delta D_{\text{test}} = \Delta D_{\text{facet}} + \varepsilon \hat{n}, \tag{10}$$

where ΔD_{facet} is a point on the facet, $\hat{n}$ is the outward normal to the facet, and $\varepsilon > 0$ is a user-specified step size. If the LMCE at the test point differs from its baseline value by more than a user-specified tolerance $\tau_{\text{LMCE}} > 0$, i.e. if

$$|\text{LMCE}_n(\Delta D_{\text{test}}) - \text{LMCE}_n(0)| > \tau_{\text{LMCE}}, \tag{11}$$

the facet is identified as a true validity boundary. Otherwise, the OPF is re-solved at the test point, the new basis polytope $\mathcal{P}_k$ is computed, and it is added to a queue of accepted polytopes to be explored. This breadth-first search continues until all facets of all accepted polytopes have been checked and no new LMCE-preserving neighbors remain. The iterative validity region is then defined as the union of all accepted polytopes,

$$\mathcal{V}^{\mathrm{I}} = \bigcup_{k\in\mathcal{K}_{\mathrm{LMCE}}} \mathcal{P}_k, \tag{12}$$

where $\mathcal{K}_{\mathrm{LMCE}}$ is the set of indices of all accepted basis regions for which the LMCE values at the flexible buses remain within the tolerance τ_{LMCE} of their baseline values. By construction, $\mathcal{V}^{\mathrm{A}} \subseteq \mathcal{V}^{\mathrm{I}}$. We assume non-degenerate OPF solutions such that each iteration crosses into a neighboring basis region. In practice, small perturbations to identical generator costs can remove degeneracy.

2.2.3. Convex Outer Approximation Since $\mathcal{V}^{\mathrm{I}}$ is a union of polytopes, it is not necessarily convex. This makes it difficult to include, e.g., as a constraint in a load-shifting optimization problem (as further described in Section 3). We therefore outer approximate $\mathcal{V}^{\mathrm{I}}$ by its convex hull $\mathrm{conv}(\mathcal{V}^{\mathrm{I}})$. To compute this, we first enumerate the vertices of each accepted polytope $\mathcal{P}_k$ from its half-space representation, then compute the convex hull of the combined vertex set. The resulting convex hull is converted back to half-space form

$$\mathrm{conv}(\mathcal{V}^{\mathrm{I}}) = \{\Delta D \in \mathbb{R}^{|\mathcal{F}|} : A\,\Delta D \leq b\}, \tag{13}$$

and enforced directly as an optimization constraint. The convex hull is an outer approximation of $\mathcal{V}^{\mathrm{I}}$, since it may include points that lie in the convex hull of accepted polytopes but outside any individual accepted polytope. However, as shown in Section 4, this approximation introduces only a small prediction error in practice, which remains negligible compared to the additional emissions reductions from larger feasible regions.

We note that for a fixed load shifting direction v, this procedure reduces to the scalar iterative validity range, where the breadth-first search over facets simplifies to checking whether the LMCE remains unchanged at the two scalar boundaries $\lambda_{\min}$ and $\lambda_{\max}$ and extending the range one basis region at a time. In this case, no convex hull approximation is needed, and the iterative validity range is exact. This convex form is required for direct inclusion in the load-shifting optimization described in Section 3.

3. Validity-Constrained Load Shifting Optimization

The computation of LMCEs and their respective validity region developed in Section 2 provides, for each operating period, a metric that maps a change in load to a change in emissions, together with the region over which that mapping remains valid. We now demonstrate how knowledge of the LMCE and validity regions can be used to inform a spatio-temporal load-shifting optimization.

3.1. Optimization Problem Formulation

For this optimization problem, let $\mathcal{F}$ denote the set of flexible nodes and $\mathcal{T}$ the scheduling horizon. The decision variable ($\Delta D_{n,t}$) represents the change in demand at flexible node $n \in \mathcal{F}$ during period $t \in \mathcal{T}$, where positive values correspond to increased demand and negative values correspond to demand reduction. Further, let $\lambda_{n,t}^{\mathrm{LMCE}}$ denote the baseline LMCE (before load shifting) at n during t. We formulate the spatio-temporal load-shifting problem as follows:

$$\min_{\Delta D} \quad \sum_{t\in\mathcal{T}}\sum_{n\in\mathcal{F}} \lambda_{n,t}^{\mathrm{LMCE}} \Delta D_{n,t} \tag{14}$$

$$\text{s.t.} \quad \sum_{t\in\mathcal{T}}\sum_{n\in\mathcal{F}} \Delta D_{n,t} = 0 \tag{15}$$

$$\bar{D}_n^{\min} \leq \Delta D_{n,t} \leq \bar{D}_n^{\max}, \quad \forall n\in\mathcal{F},\ t\in\mathcal{T} \tag{16}$$

$$-D_{n,t}^0 \leq \Delta D_{n,t}, \qquad \forall n\in\mathcal{F},\ t\in\mathcal{T} \tag{17}$$

$$\Delta D_t \in \mathcal{V}_t, \qquad \forall\, t \in \mathcal{T}, \tag{18}$$

Here, the objective (14) represents the total anticipated emission reduction from load shifting, expressed as the sum over changes in load $\Delta D_{n,t}$ multiplied by the corresponding LMCE values $\lambda_{n,t}^{\mathrm{LMCE}}$ across all nodes and time steps. Constraint (15) requires that the sum of all load shifts across nodes and time steps equals zero, which conserves the total energy consumption across the horizon and ensures that the load shifting optimization redistributes load rather than creating or eliminating it. Constraint (16) enforces that the load shifting actions adhere to the load shifting capacity defined by the lower and upper bounds $\bar{D}_n^{\min}, \bar{D}_n^{\max}$, while (17) ensures that a possible reduction in load at node n and time t cannot exceed the baseline load $D_{n,t}^0$. This latter constraint is necessary to avoid negative load. Finally, (18) represents the set of constraints that limit load shifts to remain within the validity region $\mathcal{V}_t$. Depending on the policy considered, V_t is given by the analytical validity region V_t^A, the convex outer approximation of the iterative validity region $\mathrm{conv}(V_t^I)$, or the unconstrained set $\mathbb{R}^{|F|}$.

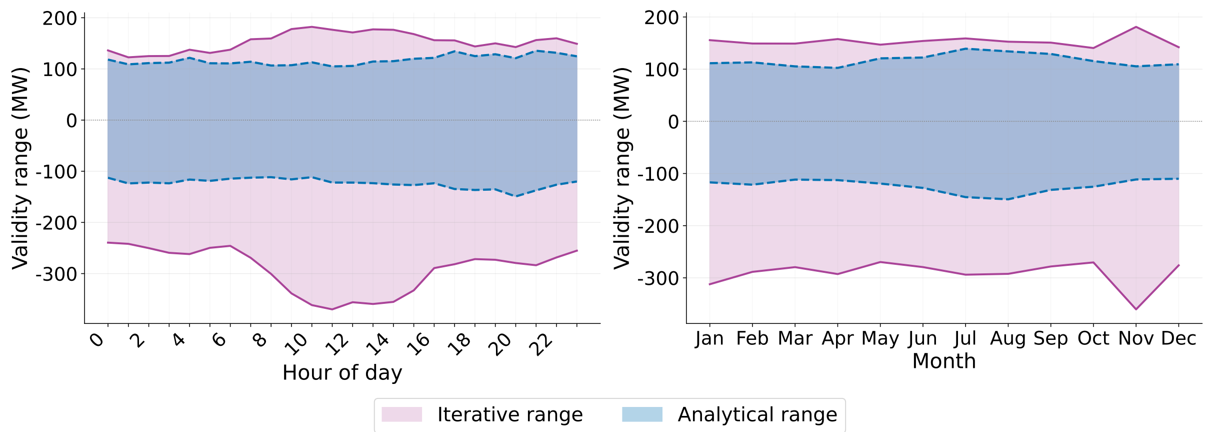

Figure 2: Mean analytical and iterative LMCE validity ranges over all buses and hours of the 2020 RTS-GMLC dataset, aggregated by hour of day (left) and month (right).

3.2. Problem Variants

We consider three problem variants with different definitions of the validity region constraints.

Iterative: In our proposed method, load shifts are constrained to remain within the convex outer approximation of the iterative validity region $\mathcal{V}^{\mathrm{I}}$, expressed as a set of linear constraints

$$A_t^I \Delta D_t \leq b_t^I. \tag{19}$$

Analytical: In this benchmark, load shifts are constrained to remain within the more conservative analytical validity region $\mathcal{V}^{\mathrm{A}}$, expressed as a set of linear constraints

$$A_t^A \Delta D_t \leq b_t^A. \tag{20}$$

Unconstrained: In this benchmarking method, we consider load shifting that is agnostic to the validity range of the LMCE. In this case, the validity region is defined as $\mathcal{V}_t = \mathbb{R}^{|\mathcal{F}|}$, meaning that load shifts are constrained only by the load shifting capacity expressed through (16), (17).

4. Results

To demonstrate the benefit of our proposed validity region computation, we conduct two sets of experiments using a full year of data for the RTS-GMLC system. First, we compare the analytical and iterative validity ranges. Second, we evaluate the benefit of integrating the extended validity regions in the carbon-aware spatio-temporal load shifting optimization.

4.1. Experimental Setup

The proposed framework is evaluated using the RTS-GMLC test system [18], which contains 73 buses, multiple conventional and renewable generators, and a full year of 2020 operating data. Using hourly generation, load, wind, and solar profiles, we generate a sequence of independent DC-OPF operating points. For each hour, we solve a baseline DC-OPF to obtain the optimal dispatch and associated LMCE values, and apply the analytical and iterative procedures of Sections 2.2.1–2.2.2 to compute the validity regions.

In all our experiments, the iterative procedure uses a step size of $\varepsilon = 1$ MW and an LMCE change tolerance of $\tau_{\mathrm{LMCE}} = 0.01\,\mathrm{kg\,CO_2/MWh}$ when checking whether the LMCE can be extended. These values are chosen to detect true changes in the marginal emissions signal while rejecting numerical noise. In our optimization experiments, each flexible bus is assigned a minimum and maximum load adjustment capacity of $\bar{D}_n^{\min} = -300$ MW, $\bar{D}_n^{\max} = 300$ MW, reflecting the aggregate flexibility of large industrial or data center loads.

All experiments were run on a MacBook Pro (Apple M3 Max, 64 GB RAM) using Julia v1.11.5 [19], JuMP v1.27.0 [20], and HiGHS.jl v1.19.0 [21] (HiGHS solver v1.14.0). Code is available upon request.

4.2. Iterative vs. Analytical Validity Regions

Single-bus validity ranges We first compare the single-bus validity ranges obtained with the iterative and analytical methods across all buses and operating hours, and characterize seasonal and diurnal variation in the size of the validity regions. The single-bus case provides an interpretable baseline for evaluating the iterative method before extending to joint load shifts. As part of this experiment, we compute analytical and iterative single-bus validity ranges over all 73 buses and 8,784 operating hours.

We observe that the validity ranges behave differently in the positive (load increasing) and negative (load decreasing) directions. For the positive direction, the iterative procedure extends the range in 13.8% to 22.2% of operating hours depending on the bus, with a median extension of 19.3% or 152 MW when an extension occurred. The negative direction exhibits a qualitatively different pattern: the iterative procedure extended it in 99.2% of operating hours, with a median extension of 117 MW. This asymmetry reflects the fact that in the downward direction, generator limits are frequently reached without changing the marginal emissions rate, causing the analytical method to systematically underestimate downward flexibility, although when extensions do occur in the positive direction, they tend to be larger in magnitude.

To discuss these results in more detail, Figure 2 shows the average validity ranges for the analytical and iterative methods, broken down by hour of day (left) and month (right). In the negative direction, we observe that the iterative validity range extends on average 250–400 MW below the analytical validity range depending on hour and season, while the two methods remain more closely aligned in the positive direction. The iterative method is most beneficial during periods of high renewable generation. Extensions are larger during midday solar hours, averaging a 32.7% increase in validity range compared to 10.6% overnight, and most pronounced during near-zero LMCE conditions, where the validity range increases by an average of 59.5% compared to 15.1% during fossil-marginal hours. Seasonal trends are consistent with this pattern, with the largest extensions occurring in November, coinciding with high wind generation.

This pattern reflects the underlying dispatch dynamics. During high-renewable periods, basis pivots frequently correspond to transitions among infra-marginal, marginal, and super-marginal generators with identical or nearly identical emissions rates. In addition, renewable generation is often distributed across many smaller units rather than a few large conventional generators, causing the marginal set to change more frequently as individual units reach their limits. Although these transitions alter the OPF basis and trigger the analytical validity boundary, they do not meaningfully change the LMCE itself. The iterative method identifies these non-consequential basis changes and continues until a genuine change in the marginal emissions signal is observed, recovering a larger region.

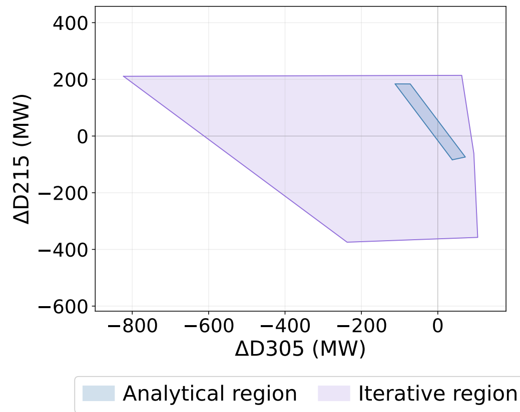

Figure 3: Analytical (blue) and iterative (purple) LMCE validity regions in the two-bus load-shift space spanned by buses 215 and 305 at hour 10 of 2020/11/07 (LMCE $=$ 0.0 gCO_2/kWh at both buses).

Since the iterative range is guaranteed to contain the analytical range, any additional flexibility it recovers represents valid operating space that the analytical method incorrectly discards. As shown in Figure 2, these extensions appear systematically throughout the year rather than at isolated operating points, with the largest deviations during high-renewable periods. Given that hours with low carbon energy are often the ones that are most valuable for load shifting, it is clear that the iterative method provides valuable additional information compared with the analytical method.

Multi-bus validity regions To understand the benefit of the iterative method in a load shifting application involving multiple buses, we next compute the validity region for a pair of buses. Figure 3 shows the analytical and iterative validity regions for buses 215 and 305 at an hour representative of near-zero LMCE conditions driven by high renewable generation. The analytical validity region (blue) is substantially smaller than the iterative validity region (purple). However, the difference between the two methods depends on the direction of the load shift. In directions where we are swapping load between the two nodes (upper left and lower right quadrant) the analytical method has a wider range, compared to directions where we are either adding or subtracting load at both nodes (upper right and lower left quadrant).

This difference occurs because net load changes, where total consumption increases or decreases, are more likely to be limited by generator limits than transmission constraints. When a generator bound is hit, the next marginal generator often has a similar emissions rate, and the iterative method can frequently recover a larger validity range. Load swaps, where consumption

Table 1: Annual load-shifting performance under unconstrained, analytical, and iterative validity constraints across three bus pairs.

Bus Pair	**Policy**	**Annual CO_2 Reduction (kt)**	**Median Daily Reduction (t/day)**	**Annual Pred. Error (kt)**	**Pred. Error (% of reduction)**	**Days Iterative > Analytical**
101 & 201	Unconstrained	443.3	1,005	994.8	69.2%	—
	Analytical	226.3	861	0	0%	—
	Iterative	258.2	1,087	0.008	0.003%	353 (96.5%)
	Improvement	*+14.1%*	*+226*	–	–	*+353*
205 & 301	Unconstrained	829.0	2,193	845.1	50.5%	—
	Analytical	231.0	825	0	0%	—
	Iterative	257.0	948	0.019	0.008%	343 (93.7%)
	Improvement	*+11.2%*	*+123*	–	–	*+343*
116 & 303	Unconstrained	650.9	1,177	1,130	63.5%	—
	Analytical	290.1	793	0	0%	—
	Iterative	325.5	889	0.049	0.015%	347 (94.8%)
	Improvement	*+12.2%*	*+96*	–	–	*+347*

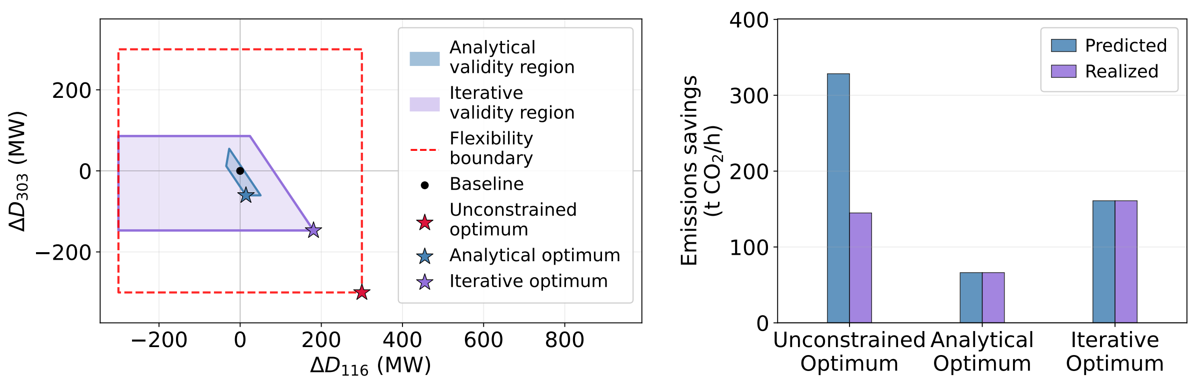

Figure 4: LMCE validity regions and policy comparison for buses 116 and 303, at hour 11 of 2020/03/03. *Left:* Two-bus load-shift space showing analytical and iterative validity regions with optimal shifts for each policy. *Right:* Predicted vs. realized emissions savings, confirming that validity-constrained policies eliminate prediction error.

shifts spatially with no net change in total demand, are more often limited by transmission constraints. Congestion often involves multiple marginal generators and is therefore more likely to change the marginal emissions rate. As a result, the iterative method leads to a smaller extension in the LMCE validity range compared to the analytical method.

4.3. Case Study: Validity-Constrained Load Shifting

We now evaluate the operational impact of iterative validity regions through a year-long spatio-temporal load-shifting study on the RTS-GMLC system. Although the proposed framework applies to arbitrary sets of flexible buses, we focus on two-bus load shifting. We consider three pairs of buses for our experiments, namely (101 $\leftrightarrow$ 201), (205 $\leftrightarrow$ 301), and (116 $\leftrightarrow$ 303), selected to represent each inter-zone transfer in the RTS-GMLC system. The optimization is solved over a 24-hour horizon, and we consider the three load shifting optimizations defined in Section 3, i.e. the **iterative** optimization problem where load shifts are constrained by $\mathcal{V}^{\mathrm{I}}$, the **analytical** optimization problem where load shifts are constrained by $\mathcal{V}^{\mathrm{A}}$ and the **unconstrained** optimization problem where load shifts are constrained only by the minimum and maximum shifting capacity of the load. After optimization, the modified load profile is applied to the RTS-GMLC system, and the DC-OPF is re-solved for every hour, enabling a true ex-post comparison of *predicted emissions savings* (i.e. the emissions savings expected based on the objective value of the load shifting problem) and *realized emissions savings* (i.e. the emissions savings obtained after resolving the OPF with the updated load profile after shifting) relative to the no-load-shifting baseline.

Table 1 summarizes the annual performance of the three approaches relative to this baseline. The unconstrained policy achieves the largest predicted CO_2 reductions (443.3, 829.0, and 650.9 kt across the three bus pairs), but incurs large prediction errors (69.2%, 50.5%, and 63.5%, respectively), meaning that over half of its apparent savings do not materialize when the system is re-simulated. This renders the unconstrained policy unreliable despite its large emission reductions.

The analytical policy eliminates prediction error entirely, at the cost of reduced flexibility. It achieves 226.3, 231.0, and 290.1 kt of annual CO_2 reductions, which are much smaller savings than what the system can support. The iterative policy improves on the analytical approach by 14.1%, 11.2%, and 12.2% across the three bus pairs, achieving annual emissions reductions of 258.2, 257.0, and 325.5 kt while maintaining prediction errors below 0.015%. Moreover, it yields larger daily emissions reductions on more than 93% of operating days, indicating that the benefit is systematic rather than confined to isolated conditions. On the remaining 3.5–6.3% of days, the two methods produce identical results.

To understand the origin of these annual improvements, Figure 4 illustrates a representative hour. The left panel shows the validity regions and optimal load shifts in the two-bus load space. The unconstrained optimum lies at the flexibility boundary, far outside both validity regions, explaining its large prediction error. The analytical optimum is confined to a narrow feasible region, achieving modest emissions reductions. The iterative optimum operates within the substantially larger region, enabling a more ambitious shift along the same direction.

The right panel highlights the key reliability tradeoff. Both validity-constrained policies achieve near-zero prediction error, whereas the unconstrained policy substantially overestimates the realized emissions savings. In this hour, the iterative method achieves a larger realized emission reduction with a smaller load shift than the unconstrained optimization.

Together, Table 1 and Figure 4 illustrate the tradeoff between flexibility and reliability. Unconstrained shifting achieves the largest emissions savings at the expense of frequent excessive shifting and an inaccurate prediction of emissions savings. In contrast, analytical shifting accurately predicts the emissions savings but unnecessarily constrains the load shifting by using an overly conservative LMCE validity region. The iterative method achieves the best of both by leveraging the larger LMCE validity region, thus recovering additional flexibility while preserving the accuracy of the predicted emissions savings.

5. Conclusion

Realizing the full potential of carbon-aware flexibility will require not only computing marginal carbon signals, but understanding the regions over which those signals can be trusted. In this paper, we propose a framework to compute LMCE validity regions for multi-node load shifts that extend beyond the regions identified through simple LP (linear programming) sensitivity analysis. We then embed the LMCE validity regions as constraints in a spatio-temporal load-shifting optimization. Given that the LMCE validity regions are the union of several polytopes and generally non-convex, we outer approximate them using their convex hull to obtain a convex optimization problem. Finally, we demonstrate the use of the LMCE validity regions in a year-long case study on the RTS-GMLC system. In this context, the present work yields three main findings:

First, we demonstrate that our iterative approach provides substantial benefits over analytical methods that define the LMCE validity regions solely based on LP analysis. Analytical methods can be overly conservative because basis changes in the OPF solution do not necessarily imply changes in the marginal carbon signal. Importantly, this effect is particularly significant during periods of high renewable generation, when load shifting is most likely to happen.

Second, we demonstrate that the proposed validity regions accurately describe the multidimensional load shifting region whose shape reflects the coupling between nodes through shared generation and transmission constraints. These couplings introduce a directional dependence that scalar LMCE validity ranges derived for load changes at individual nodes cannot capture.

Third, we show that validity-constrained load shifting with our proposed iterative LMCE validity regions achieve 11–14% greater annual emissions reductions compared to the analytical region, while significantly reducing prediction errors and excessive shifting compared with unconstrained load optimization. The inaccuracy introduced by using the convex hull outer approximation to represent the validity region is very small, accounting for only 0.015% of total predicted emissions savings.

These findings have important implications for carbon-aware demand flexibility. The practical value of LMCE-guided load shifting depends critically on knowing how far loads can shift while maintaining accuracy of the predicted emissions reductions. Accurately characterizing this region in both *shape* and *size* is what enables validity-constrained load shifting to be simultaneously reliable and effective. More broadly, these results challenge the conventional view that LMCEs are fragile signals with limited practical validity. The carbon signal is more robust than scalar, basis-preserving analysis implies, and that robustness is most pronounced exactly when and where it matters most for decarbonization.

Several limitations point toward future work. While the framework conceptually supports arbitrary sets of flexible nodes, the case study considers only two-dimensional load-shift spaces. Increasing to higher-dimensional load shifting increases the computational cost of the iterative procedure, which needs to solve an additional OPF problem for each neighboring basis region until a true LMCE change is detected. A related limitation is that embedding V^I as an optimization constraint requires approximating it by its convex hull, which is an outer approximation with no theoretical bound on the gap and is computationally demanding to compute in higher dimensions. Future work could explore other options for representing the union of polytopes, using e.g. integer programming and SOS1 constraints. Additionally, while the DC-OPF formulation adopted here is the appropriate model for targeting real-time market dispatch, it neglects reactive power flows, voltage constraints, and unit commitment decisions that may further restrict validity regions in practice. Extending validity analysis to AC power flow or security-constrained settings, and to stochastic or robust formulations that account for forecast uncertainty in load and renewable generation, are important directions for next steps.

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